Worksheet 2

Measures of central tendency

Complete exercises

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 25 min

Exercise 1 – Choosing and calculating the appropriate measure of centre

1. Basket values are $18, 20, 20, 22, 25, 95$ euros. Calculate mode, median and arithmetic mean. Which measure best represents a typical customer? Why can the mean still matter?
2. Three grades are 14, 11 and 16, weighted by 30%, 30% and 40%. Calculate the weighted mean.
3. Revenue changes by $+5\%$, -2% and $+12\%$. Calculate the appropriate average annual growth rate.
4. A vehicle travels 120 km at 60 km/h and then 120 km at 90 km/h. Calculate the appropriate average speed.

Core application 10 min

Exercise 3 – Which indicator should be used?

For each situation, choose the most appropriate indicator and justify in one sentence: most frequently sold product category; typical salary with a few very highly paid executives; average price when purchased quantities differ; average multi-year growth; average speed over equal distances; centre of a roughly symmetric numerical distribution without extreme values.

Core application 15 min

Extra 1 — Robust centre for delivery performance

Delivery times (minutes) for 12 orders are: 18, 19, 20, 20, 21, 21, 22, 22, 23, 24, 25, 65.

1. Calculate the mode, median and mean.

2. Recalculate the mean without the 65-minute delivery.
3. Recommend the principal indicator for a customer-facing dashboard and justify your choice.

Compare the indicators before and after removing the unusual observation; do not delete it without explanation.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 30 min

Exercise 2 – Grouped distribution

Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5

1. Identify the modal class, taking unequal widths into account.
2. Calculate the median by linear interpolation.
3. Calculate Q_1 and Q_3 by interpolation.
4. Calculate the approximate mean using class midpoints.
5. Find the maximum price such that about 60% of customers can afford at least that price.

Assessment / exam practice 20 min

Extra 2 — Integrated centre-selection case

A retailer reports three indicators:

Question	Data
Average basket value	120 baskets: €3,600 total
Typical annual growth	+12%, −8%, +15%, +5%
Average speed	Two equal 90-km legs at 60 and 90 km/h

1. Choose and calculate the appropriate mean for each row.
2. Explain why using the arithmetic mean in all three cases is incorrect.
3. Write a short managerial conclusion.

Match additive totals, compounded factors and equal-distance rates with arithmetic, geometric and harmonic means.